
CE 617: Mechanics of Composite Materials

Course Code: CE 617

Course Name: Mechanics of Composite Materials

L-T-P-C: 3-0-0-3

Prerequisites: Mechanics of Solids

Students intended for: MS/PhD

Elective or Core: Elective

Mutual Exclusion: ME 617

Approval: 12th Senate (ME 617), 59th BoA (CE 617)

Course contents

- **Module I:** Introduction, Classification of Composite Materials, advantage of composite materials and applications [4 Lectures]
- **Module II:** Macro-mechanical behaviour of a lamina; Stress strain relationships for anisotropic materials, stiffnesses and compliances, restrictions of engineering constants, invariant properties of lamina, Strength of lamina [10 Lectures]
- **Module III:** Micromechanical behaviour of lamina: Mechanics of materials approach for stiffness, elasticity approach, imperial approaches, strength estimations [12 Lectures]
- **Module IV:** Micromechanical behaviour of Laminates: Classical laminate theory, special cases of laminates, strength of laminates, Interlaminar stresses [12 Lectures]
- **Module V**
Experimental characterization of composites: Uniaxial, biaxial tension and compression tests, interfacial and inter laminate shear stress characterizations. [4 Lectures]

Text and Reference Books:

1. Laszolo P Kollar and George S. Springer, **Mechanics of Composite Structures**, Cambridge university Press, 2003.
2. J. N. Reddy, **Mechanics of Laminated Composite Plates and Shells Theory and Analysis**